

Leibniz’s law of continuity, read on records, selects a positive action floor for motions

Result, 2026-09-29 (Claude; written derivation and passage readings; refereed by Fable with REFINE, corrections applied). In the mark model of the Planck paper (§3), the infimum of the error with which a non-adaptive protocol, read by tests invariant under the unknown initial position and velocity, decides Newton’s comparison, inertia against a constant force F over a cell of duration τ , is

$$P_*(F) = \Phi\left(-\frac{1}{2}\sqrt{K_\tau/\kappa}\right), \quad K_\tau = \frac{F^2\tau^3}{24m},$$

for a mark floor $\kappa > 0$, while $P_*(F) = 0$ for every $F \neq 0$ when $\kappa = 0$; at $F = 0$ the cases coincide and $P_* = \frac{1}{2}$. The infimum is approached by dense protocols. **The best verdict over these protocols is continuous in the force at $F = 0$ if and only if $\kappa > 0$;** each single protocol gives a continuous verdict on both branches. The same holds on Zeno’s rung, rest against uniform motion, with $mv^2\tau/(8\kappa)$ in place of $K_\tau/(4\kappa)$, attained by two marks. A single cut at fraction s carries exactly the discernibility $d^2 = 3s(1-s)K_\tau/\kappa$, the cut measure’s share (§3). Static shapes behave differently: when the read coordinate has no dynamics in the window, the best verdict jumps at zero for every κ . The equivalence concerns the observer ignorant of the preparation; a protocol that knows it (Yuen’s, in §3 of the paper) decides at every force for every κ .

Leibniz published the law of continuity in July 1687, the year of the *Principia*, as a test of laws of motion: when two cases approach and are lost in one another, their outcomes must do the same. He measures the difference of the cases by discernibility, describing the nearly equal case of a collision as one “dont à peine ce cas peut estre distingué” (§2). Applying the same measure to the outcomes is our step. Read that way, the law holds for Newton’s comparison exactly when $\kappa > 0$. Read with outcomes measured by positions and velocities, as Leibniz applied it, it holds on both branches. The fork that every route of this programme reaches (positivity as a premise, the zero branch admissible) is therefore located between a reading Leibniz made and one he did not, and the programme’s thesis, that quantization is a consistency condition on laws which Newton’s continuum limit missed, has a period form: Leibniz’s “preuve ou examen” of laws of motion, applied to their records. His *petites perceptions* (1704) supply the graded, additive discernibility that the records reading needs (§3), and his *Pacidius* (1676) the proportion argument against leaps, which a floor on action meets (§2b). The Newton-age dispute over indivisibles, Galileo, Cavalieri and Guldin between 1621 and 1647, concerned static figures, where the theorem imposes no floor and Newton’s limit doctrine is the right answer (§4).

After refereeing (Claude Fable, 2026-09-29). REFINE, applied. The mathematics of Proposition L checked, including the $\kappa = 0$ case and the static case; its equivalence at $F = 0$ needs only Theorem 2’s inequality. Fable supplied the exact single-cut share $3s(1-s)K_\tau/\kappa$, now in §1 and §3. Corrections: the non-adaptive, preparation-ignorant hypotheses are stated up front (Yuen’s protocol jumps for every κ); the transfer of Leibniz’s discernibility from the cases to the outcomes, the reading of his gunpowder exemption (with its counter-reading) and the confinement to motions are marked as ours; “distingué” and “sçauroit” are flagged as reconstructed from the OCR; the Epicurus, al-Nazzam and Galileo wordings of §4 are softened to what the texts say; prior art on statistical distance and on continuity axioms is added.

1. The proposition

Setting. The statistical model of the Planck paper, §3: mark j at time t_j returns the position with Gaussian error of standard deviation δ_j and delivers an independent Gaussian impulse of standard deviation Δ_j ; every available mark obeys $\delta_j\Delta_j \geq \kappa$; protocols are non-adaptive and

tests are invariant under the unknown y_0, v_0 . The optimal invariant test errs with probability $\Phi(-d/2)$, and Theorem 2 of the paper gives $d^2 \leq K_\tau/\kappa$ for every protocol, with a sharp constant.

Proposition L. Let $P_*(F)$ be the infimum of the error over admissible protocols and invariant tests. Then $P_*(F) = \Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa})$ if $\kappa > 0$, and $P_*(F) = 0$ for $F \neq 0$ if $\kappa = 0$. For Zeno's rung (rest against speed v , unknown initial position), $P_*(v) = \Phi(-\sqrt{mv^2\tau/(8\kappa)})$ if $\kappa > 0$ and $P_*(v) = 0$ for $v \neq 0$ if $\kappa = 0$. For a static taper ϑ read at two heights, $P_*(\vartheta) = 0$ for $\vartheta \neq 0$ and every $\kappa \geq 0$, when the read coordinate has no dynamics in the window (a clamped figure, or one of effectively infinite mass). For a single cut, marks at $0, s\tau, \tau$, the best d^2 is exactly $3s(1-s)K_\tau/\kappa$.

Proof. For $\kappa > 0$, $\Phi(-d/2)$ decreases in d , and the supremum of d^2 over protocols is K_τ/κ by the sharpness in Theorem 2; hence the infimum of the error. For $\kappa = 0$, marks with $\Delta_j = 0$ and arbitrary $\delta_j > 0$ are admissible; take three marks at distinct times and weights u with $\sum u_i = \sum u_i t_i = 0$, so that $u^\top P = \frac{F}{2m} \sum u_i t_i^2 \neq 0$ while $u^\top \Sigma u = \sum \delta_i^2 u_i^2 \rightarrow 0$; then $d \rightarrow \infty$ (the same holds at fixed δ by repetition, since $\Delta = 0$ turns the motion into a static figure). The equivalence at $F = 0$ needs only Theorem 2's inequality, whose proof is complete: $d^2 \leq K_\tau/\kappa$ gives $P_* \geq \Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa}) \rightarrow \frac{1}{2}$; the sharpness fixes only the exact value. Zeno's rung is Theorem 9(i) of the paper, whose bound $d^2 \leq mv^2\tau/(2\kappa)$ two marks attain. The static case is Theorem 9(iii): with N marks of resolution δ at each height, $d^2 = N\vartheta^2\Delta h^2/(2\delta^2)$, unbounded in N whatever κ is. For one cut, the constraints force $u \propto (-(1-s), 1, -s)$; then $S_0 = S_2 = 0$ and $S_1 = -s(1-s)\tau u_1$, so the noise is at least $\delta_1^2 u_1^2 + \Delta_1^2 S_1^2/m^2 \geq (2\kappa/m)s(1-s)\tau u_1^2$ by the arithmetic-geometric mean, with equality for sharp end marks (their kicks do not propagate) and $\delta_1 = \Delta_1 s(1-s)\tau/m$; the signal is $(F\tau^2/2m)^2 s^2(1-s)^2 u_1^2$, and the ratio is $F^2\tau^3 s(1-s)/(8m\kappa) = 3s(1-s)K_\tau/\kappa$. $\square$

For $\kappa > 0$ the verdict is Lipschitz at $F = 0$ with constant $\tau^{3/2}/(2\sqrt{2\pi}\sqrt{24m\kappa})$, which diverges as $\kappa \rightarrow 0$: the zero branch is the singular limit of a family of continuous verdict functions, the pattern of an order of limits. Here N counts marks on one body inside one window, Newton's single trajectory; repeating the whole experiment on fresh preparations multiplies d^2 on either kind of case. The paper's quantum results (§§4–6) reproduce the split for the observer ignorant of the preparation: a static parameter can be estimated without limit by repeated marks, while the dynamical comparison in a fixed window keeps the N -independent bound. With knowledge of the preparation (Yuen's protocol) a single sharp mark decides at every force, so the proposition is a statement about the preparation-ignorant observer.

2. Leibniz, 1687: the law of continuity as a test of laws

Leibniz, “Lettre de M. L. sur un principe general utile à l'explication des loix de la nature”, *Nouvelles de la République des Lettres*, July 1687, in Gerhardt, *Die philosophischen Schriften* III (1887), 51–55 (companion; passage, OCR normalized by hand; where the OCR misreads accents, the reconstructed letters are flagged).

- **The text.** “Lorsque la difference de deux cas peut estre diminuée au dessous de toute grandeur donnée *in datis* ou dans ce qui est posé, il faut qu'elle se puisse trouver aussi diminuée au dessous de toute grandeur donnée *in quaesitis* ou dans ce qui en resulte ... *Datis ordinatis etiam quaesita sunt ordinata*” (p. 52). “Le repos peut estre considéré comme une vistesse infiniment petite, ou comme une tardité infinie”, so that the rule of rest must be a particular case of the rule of motion, “autrement ... ce sera une marque assurée, que les regles sont mal concertées” (pp. 52–53). Against Descartes's second rule of collision, after “une augmentation aussi petite que l'on voudra du corps B auparavant égal à C”, B should reflect a little less and C a little more “qu'au cas de l'égalité dont à peine ce cas peut estre distingué” (p. 53; the OCR reads “distinguo”, and the final letter awaits the page image). And the exemption: “dans les choses composées quelques fois un petit changement peut faire un grand effect, comme par exemple une estincelle tombant

dans une grande masse de la poudre à canon”, while “à l’égard des principes ou choses simples, rien de semblable ne scauroit arriver” (p. 54; OCR “scauroit”).

- **What it commits him to.** A law of motion is tested from outside, before any inner discussion (“preuve ou examen . . . avant même que de venir à une discussion interieure”), by whether its outcomes converge when its cases do. Rest is the limit of slow motion and equality the limit of inequality, and a rule that jumps at either is ill made. Composite amplifiers may magnify small differences into large effects; simple principles may not jump.
- **What the theorem does with it.** Leibniz applies the test to collision velocities, the geometric reading, and Newton’s laws pass it on both branches. His phrase measures the difference *in datis* by discernibility: the nearly equal collision can scarcely be told from the equal one. The records reading applies the same measure *in quaesitis*, and that transfer is ours. Proposition L shows what the test then demands: for Newton’s comparison and for Zeno’s rung, continuity of the best verdict holds exactly when $\kappa > 0$. The exemption marks the boundary of the argument. An instrument is a composite amplifier, and finite amplification is allowed; the zero branch needs the supremum over all instruments the laws permit to jump, a jump in what the laws allow to be recorded, which is the level of principle if the mark trade-off is a law. That mapping of Leibniz’s line between composites and simple principles onto the line between one instrument and the laws’ supremum is ours, and it has a counter-reading: Leibniz exempts effects for which “on en peut rendre raison par les principes generaux mêmes” (p. 54), and a Leibnizian can explain the $\kappa = 0$ jump from $\kappa = 0$ mechanics by ever finer instruments. Whether the verdict of the best possible record counts among “ce qui en resulte” is the premise, now stated in Leibniz’s terms. The theorem also confines the reading: for static shapes the records reading fails on both branches. Leibniz applies the law to geometry and physics alike (“absolument necessaire dans la Geometrie, mais il reussit encor dans la physique”, p. 52), so the confinement to motions is a physical choice of ours, resting on back-action; its textual defence is that Leibniz uses the law as a test of laws of nature, and a taper is a datum rather than a law.

Prior art, at metadata level (Crossref verified 2026-09-29 where a DOI is given): Mortensen, “The Leibniz Continuity Condition, Inconsistency and Quantum Dynamics”, *J. Philos. Logic* **26** (1997), 377–389, treats the condition against quantum jumps at the instant of change; the pairing of continuity with the identity of indiscernibles is Chapter V of Russell, *A Critical Exposition of the Philosophy of Leibniz* (1900). Leibniz repeats the test against Descartes’s rules of collision in the *Animadversiones in partem generalem Principiorum Cartesianorum* (1692) and the *Specimen dynamicum* (1695), so the 1687 “preuve ou examen” is a settled method (recalled, not read here). The condition used here is continuity of the parametrized family of recorded states in statistical distance (Wootters 1981; Braunstein and Caves 1994), in which classical pure states are pairwise perfectly distinguishable. It is weaker than Hardy’s continuity axiom, which asks for a continuous reversible transformation between pure states (see the Hardy audit); continuity axioms separate classical from quantum theory also in Masanes and Müller 2011 and Chiribella, D’Ariano and Perinotti 2011.

2b. Leibniz, 1676: little rests, leaps and the animalcula

Leibniz, *Pacidius Philalethi. Prima de motu philosophia*, headed “Scripta in navi qua ex Anglia in Hollandiam trajeci. 1676 Octob.”, in Couturat, *Opuscules et fragments inédits* (1903), 594–627 (companion; passage, OCR normalized by hand).

- **The text.** “Ex veteribus Empedocles et ex recentioribus docti quidam Viri quietulas quasdam interspersas asseruère” (p. 605), because they “capere non potuerunt quomodo motus unus alio celerior esse possit, sine quiete interspersa” (p. 606). Charinus concedes the regress: “tametsi indefinite progrededer subdividendo ac quietulas indefinite exiguas

atque indesignabiles, motulis ejusdem naturae miscerem, opus tamen et tempusculis atque lineolis foret” (p. 606), and a rotating radius gives unequal speeds without any rest. A leap is transcreation: the body is “extingui et annihilari, et in B momento post iterum emergere ac recreari” (p. 617). Against it, “cùm enim magnitudo aut parvitas nihil ad rem faciat”, animalcula as much smaller than us as a head is than the earth would find the same absurdity in their leaps, “omnia proportione sibi respondent”, so the leaps are “semper ad minora ac minora propelli et nusquam consistere posse in natura rerum ... nulla autem ratio est, cur huic potius quàm illi corpusculorum gradui saltus illi miraculosi ascribantur, nisi atomos scilicet admittamus” (pp. 617–618). Leibniz denies such atoms by the same argument.

- **What it commits him to.** Speed differences need no rests; rests mixed with motions either concede continuous motion or require leaps; a leap of any size needs a privileged grade of bodies, which sufficient reason refuses; hence no leaps, eleven years before the law of continuity.
- **What the theorem does with it.** The little rests are the velocity-switching processes of the telegraph and checkerboard notes: motion at one speed whose state switches at a rate $\omega = mc^2/K$ (there between the two directions; between motion and rest the structure is the same), with the mean speed set by the mixture. Charinus’s regress, rests made indefinitely small among little motions, is the limit $K \rightarrow 0$, which converges to continuous motion at the mean speed; the regress is a limit, and its end is the zero branch. The radius answers the motive for the rests and leaves the rate open. Transcreation below the mesh is the paper’s reading of al-Nazzam’s leap. The animalcula argument is sound against a leap of fixed size: Newton’s mechanics has the two-parameter similarity $x \rightarrow \lambda x$, $t \rightarrow \mu t$ (the dilations of the fifth-postulate note, the zoom of the tangent-groupoid note), and a length breaks it. A floor on action fixes no length. Under $x \rightarrow \lambda x$, $t \rightarrow \lambda^2 t$ at fixed mass, $F \rightarrow F/\lambda^3$ and $K_\tau = F^2 \tau^3 / (24m)$ is invariant, as is $mv^2 \tau$, so animalcula scaled this way reach the same verdicts in the corresponding cases: “omnia proportione sibi respondent” survives on a one-parameter subgroup, the diffusive scaling of quantum paths. The dilemma between atoms and no leaps omits this option. What sufficient reason then asks for is a reason for an action scale. The composition results give one constant for all bodies (rotation composition; $\kappa = mD$ in the stochastic route), so that scale converts units of mass, length and time and its value is a convention. The question sufficient reason leaves is zero or positive, which §1 settles for motions under the records reading of the 1687 law. Leibniz between 1676 and 1704 thus supplies the objection and the reply: proportion forbids a privileged size, which a floor on action does not need, and continuity of outcomes read on discernibility requires the floor.

3. Leibniz, 1704: small perceptions add

Leibniz, *Nouveaux essais*, Préface, written 1703–1705, in Gerhardt V (1882), 47–49 (companion; passage, OCR normalized by hand).

- **The text.** To hear the roar of the sea “il faut bien qu’on entende les parties qui composent ce tout, c’est à dire les bruits de chaque vague ... Car il faut qu’on en soit affecté un peu par le mouvement de cette vague ... autrement on n’auroit pas celle de cent mille vagues, puisque cent mille riens ne sauroient faire quelque chose” (p. 47). “La nature ne fait jamais des sauts ... jamais un mouvement ne naist immédiatement du repos ny s’y réduit que par un mouvement plus petit” (p. 49). “En vertu des variations insensibles, deux choses individuelles ne sauroient estre parfaitement semblables” (p. 49).
- **What it commits him to.** Every effect, however small, registers a little; perception is graded and additive; noticeable perceptions come by degrees from insensible ones; distinct individuals always differ, by differences that can be insensible.

- **What the theorem does with it.** The mark model makes the addition exact. A single cut at fraction s carries the discernibility $3s(1-s)K_\tau/\kappa$ (Proposition L), which is the share of Newton’s cell action that the cut measure assigns to that cut (Theorems 1–2), and nested cuts add their shares toward K_τ/κ in the balanced dense limit (the orthogonal hats of Theorem 2 there; Proposition 4 gives the same decomposition as Cameron–Martin energy). The doctrine has two halves, and they fall differently. “Cent mille riens ne sauroient faire quelque chose” asks that every share be positive, which holds for every $\kappa \geq 0$. “Insensible alone” asks that a share be unable to decide by itself, and that half selects the floor: with $\kappa > 0$ a cut shorter than the mark mesh cannot exhibit the force alone and contributes in the assembly, while with $\kappa = 0$ a single cut decides and records of motion have no insensible parts. The zero branch can still place insensibility in the perceiver, as Leibniz’s confusion of created substances does, graded from one substance to another; the stochastic route shows how composition turns graded constants into one ($\kappa = mD$ for every body), and the step from confused perceivers to one universal floor is recorded here as a reading. The maxim on motion from rest is Galileo’s passage through all degrees of slowness, stated at the beginning of motion, the regime of Lemma X (“ipso motus initio”); Zeno’s rung of Proposition L is its records version.

4. The Newton-age dispute over indivisibles, 1621–1647

Book I’s closing scholium prefers limits to indivisibles, “quoniam durior est indivisibilium Hypothesis”, without naming the dispute it settles. The primary texts are held in `docs/classics`.

- **Galileo’s bowl and cone** (*Discorsi*, Giornata prima, Favaro VIII, 74–76, 78–80; companion). Every horizontal section of the bowl (cylinder minus hemisphere) equals the section of the inscribed cone, so the surfaces are “sempre eguali, e . . . diminuendosi sempre egualmente, vadano a terminare l’una in un sol punto e l’altra nella circonferenza d’un cerchio . . . perché in questa conseguenza sola versa la nostra maraviglia”. His conclusion: “questi attributi di maggioranza, minorità ed egualità non convenghino a gl’infiniti”. *Commitment*: comparison fails among infinites and indivisibles. *The theorem*: the bowl is Democritus’s third (bowl and cone are each a third of the cylinder), a static figure, so Theorem 9(iii) and Proposition L put no floor on it, and Newton’s scholium to Lemma XI declines the question that makes the wonder: ultimate ratios are limits of ratios, never ratios of ultimate quantities, so the ratio of sections is 1 at every cut and point and circle are never compared. Leibniz takes the opposite side from Galileo in 1687, without engaging his paradox: equality is an infinitely small inequality, so the attributes extend to the limiting case by continuity.
- **Cavalieri’s reply** (letter 2992 of 1634, Opere XVI 136–138; companion). Equal parts are removed “essendo noi arrivati al nullo piano tanto nel cono quanto nella scodella”, and “non mi dichiaro di componere il continuo d’indivisibili, ma solo mostro che i continui hanno la proportionione delli aggregati di questi indivisibili”. *Commitment*: ratios of aggregates of sections, with no claim of composition. *The theorem*: the cut measure keeps exactly this stance, a statement about aggregates of cuts (shares summing to K_τ for every schedule); it composes no continuum, and the mark mesh bounds exhibition while leaving division free.
- **Galileo’s wheel** (Favaro VIII, 69–72, same companion). A smaller concentric wheel covers the larger wheel’s line with its own sides “con l’interposizione di cento mila spazii vacui traposti”, and for circles “sì come i lati non son quanti, ma bene infiniti, così gl’interposti vacui non son quanti, ma infiniti”; Simplicio hears in it “quei vacui disseminati di certo filosofo antico”, and Salviati’s retort about the one “il quale negava la Provvidenza divina” points to Epicurus without naming him. *Commitment*: a continuum composed of infinitely many unquantified indivisibles, partly full and partly void, crossed by leaps across the

voids. *The theorem*: in 1638 Galileo publishes as consistent a structure that converges with al-Nazzam’s leap; no transmission evidence is known, and the entry records convergence. The paper’s reading of the leap applies: below the mesh a record is consistent with a leap, the interval stays divisible, and the crossing is exhibited in finitely many steps.

- **Guldin** (*Centrobaryca* IV, 1641, preface p. 4; companion): “Galileus profecto in eodem Dialogo de Motu locali, disputans de infinito, de proprietatibus finitorum, quas infinitis applicare minime liceat, contra ipsum concludit.” (The companion’s transcription is read from the page images by one reader.) *Commitment*: Galileo’s own principle refutes Cavalieri. *The theorem*: it sides with Guldin and Newton on composition (no least magnitude follows from κ) and with Cavalieri on ratios of aggregates.

The four entries share one feature that sharpens the paper’s §9. The seventeenth-century dispute Newton answered concerned static figures, and there the theorem agrees with his answer and imposes nothing. The floor enters only where a record carries back-action, in the lemmas applied to motion, and the dispute about motion is the one he passed over.

5. Consequence for STATE

For the Newton goal: for the observer ignorant of the preparation, using non-adaptive protocols, positivity of the action floor is equivalent in the mark model to the records reading of Leibniz’s 1687 law of continuity for laws of motion (Proposition L). The zero branch keeps the geometric reading, which Leibniz himself applied, so the fork stands; it now lies between a reading Leibniz made and one he did not, with his measure of the cases, “à peine distingué”, and the “insensible alone” half of his doctrine of small perceptions on the side of the floor, and statics showing why the reading must be confined to motions, by a physical choice of ours. A single cut carries exactly the cut measure’s share of the discernibility, a result of the mark model worth citing on its own. For the scholion: three Leibniz entries (1676, 1687, 1704) and the Galileo–Cavalieri–Guldin layer are supplied with their three obligations; the paper’s §9 carries a pointer. §§1–4 are refereed (Fable, REFINE, applied); §2b awaits a referee.